Erwhan Le Couëffec

R&D Engineer specialised in photonics — I always test theory against measurement

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French national (EU free movement, no sponsorship needed) | Driving licence B | Available immediately

PROFILE

Generalist engineer trained at CentraleSupélec (programme completed in 2026), specialised in photonics, with 4 years of apprenticeship experience in industry and R&D: optical metrology, 3D measurement, calibration and instrumentation. I enjoy understanding a system in depth, measuring it and improving it through rigorous experimental work. I am looking for an R&D engineering role with a strong hands-on component, preferably in the Basel area.

PROFESSIONAL EXPERIENCE

R&D Engineer (apprenticeship) 10.2025 – 09.2026
Holo3 — Saint-Louis (FR), Basel area · optical metrology

- Mission 1 · Calibration-target detection: preliminary study on 3 targets (~1,300 measurements), then introduced Design of Experiments to the team on my own initiative (216 -> 27, then 432 -> 32 configurations), cutting the number of trials and quantifying each parameter's influence: all act on black/white contrast.
- Mission 2 · HoloScan 3D system calibration (thesis): theory and uncertainty propagation, then 5 comparative studies (fringe projection, 2 cameras + 1 projector), validated on a certified 250.1097 mm ball bar: ±120 µm criterion met in 108 of 110 measurements, repeatability < 2 µm.
- Mission 3 · Vision measuring machines, in parallel: Mitutoyo qualified on a reference target (repeatability ~1 µm), compared with a Hexagon (target curvature 93 vs 155 µm); target characterisation identified as the main lever for calibration quality.

WCM / Continuous Improvement Engineer (apprenticeship) 10.2023 – 06.2025
Placoplatre (Saint-Gobain group) — Chambéry (FR) · plasterboard manufacturing

- Plaster viscosity check: designed an OK/NOK fixture that sets the position of the ruler used to measure the spread diameter, turning an operator-dependent measurement into a repeatable one.
- Autonomous maintenance across the whole plasterboard line: mapped every machine with a risk-zone score (machine criticality × area cleanliness × ease of access) to prioritise actions.
- Industrial risk analysis: risk × accident-probability scoring, assessed without and with PPE, to quantify the gain of each improvement (e.g. a raised safety barrier preventing tipping).
- 5S project in the production tool storage area; cross-functional work with production, maintenance and quality.

Measurement / Industrialisation Apprentice 09.2021 – 08.2022
Wilo France — Laval (FR) · pumps for sanitary and fire-protection systems

- Created leak testing, previously non-existent, for 16 HomeBooster booster-pump models, grouped into 3 parameterised ATEQ programs (1 min 30 s test, leak criterion <= 6 cm³/min): simpler choice for operators, about 30 s saved per test.
- EPA zone (Electrostatic Protected Area) on the Plavis line: defined the zone, measured the resistivity and grounding of its elements; line inspected by an external body as part of the site certification.
- Inspection protocol for electrical cable crimping; compiled the welding parameters of all machines.

EDUCATION

CentraleSupélec generalist engineering programme, apprenticeship track 2023 – 2026
CentraleSupélec, Metz (FR)

Final year: Photonics and Nano-systems Engineering. Thesis: HoloScan system calibration. Programme completed; the degree (Master's level, EQF 7) is awarded once the required C1 English certification is validated.

Classe préparatoire ATS (intensive maths & physics year) 2022 – 2023
Lycée Joliot-Curie, Rennes (FR)

Intensive maths & physics year (~30 students): 2nd in the first semester, 1st in maths and physics; top 10 over the year.

DUT Mesures Physiques (2-year technical degree, 120 ECTS) 2020 – 2022
Université du Mans (FR)

Materials and measurement systems. Year 1: top 15 of ~80. Year 2 as an apprentice at Wilo France: top 3 of ~12 apprentices.

ACADEMIC PROJECTS (SELECTION)

- 1550 nm DFB laser diode: L-I curve, optical spectrum (53 dB side-mode suppression), relaxation oscillations up to 8.5 GHz and optical feedback.
- Free-space Raman spectroscopy: backscattering setup built from scratch, identification of 8 unknown plastics.
- Holography: recording and reconstructing the 3D hologram of a metal part at 532 nm.
- Photonics simulation: beam propagation in a photorefractive crystal (Kukhtarev model, BPM), optoelectronic reservoir computing, Monte Carlo methods.

SKILLS

Metrology & R&D	Optical metrology, fringe-projection 3D measurement, calibration, design of experiments, uncertainty analysis, characterisation (trueness, precision), industrial cameras
Optics & lab	Optical benches, cameras and projectors, lasers (DFB, He-Ne, Nd:YAG), Raman spectroscopy, optical and RF spectrum analysers, polarisers, half- and quarter-wave plates, oscilloscope
Industry & quality	Leak testing (ATEQ), EPA zones, 5S, autonomous maintenance, risk analysis
Software	MATLAB, advanced Excel, Python (basic: simulation and data-processing scripts), LTspice, GOM Inspect, SensorPhase, CalibTool; previously LabVIEW, SolidWorks

LANGUAGES & INTERESTS

French: native | English: B1/B2 | German: beginner (A1), lessons planned | Chinese: basic (A1/A2)

Handball, bouldering, following sport (especially NBA basketball) | References available on request